Date: 16/10/26.

CN - Assignment 02
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Roll-no: 24K-0737.

Task #1:-

1.
Ans: Reliable data transfer ensures casually lists, supply requests, and review assignments reach their destination correctly & in order. It detects corrupted or lost packets & retransmits missing data, preventing incorrect information.

2.
Ans: Stop-&-wait sends one packet & waits for its acknowledgement before sending the next. Here, transmitting an 8000 bit packet takes only 4ms, while the RTT is 500ms. Therefore, most time is wasted waiting, resulting in low link utilization.

3.
Ans: Pipelining allows multiple packets to be sent before acknowledgements arrive. With a window size of four, up to four packets can be transmitted without waiting for individual acknowledgements, improving bandwidth utilization.

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5.

Ans: In this scenario, Selective Repeat (SR) is more suitable because:

(i) Efficient retransmission:

(ii) Out of order buffering.

=) SR requires more receiver memory & more complex acknowledgement & timer management than GBN.

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2) Task #2.

Given:

Total data packets = 10
Sender window size N = 4
packet 2 is lost
packet 3 & 4 reach the receiver

(a) Go-Back-N (GBN):-

Event	Sender packet/Action	Receiver/Ack	Sender window
1	send pkt 0	Receive pkt 0, sends 0	[0-3]
2	Send pkt 1	Receive pkt 1, send Ack 1	[0-3]
3	Send pkt 2	Pkt 2 is lost	[0-3]
4	send pkt 3	Discard pkt 3, duplicate Ack 1	[0-3]
5	Ack 0 received	Sender advances base to 1	[1-4]
6	Send pkt 4	Discards pkt 4 if pkt 2 is still missing	[1-4]
7	Ack 1 received	Sender advances base to 2	[2-5]
8	Timeout for pkt 2	Retransmit outstanding pkts 2, 3 & 4	[2-5]
9	Receive pkt 2	Accept pkt 2, sends Ack 2	[2-5]
10	Receive pkt 3	Accept pkt 3, sends Ack 3	[2-5]
11	Receive pkt 4	Accept pkt 4, send Ack 4	[2-5]
12	Acks received	sender advances its window	[5-8]
13	Send pkts 5, 6, 7 & 8 as permitted	Receive in order	Sliding.
14	send pkt 9 when permitted	Receives pkt 9 & sends 9	Sliding.
15	Final Ack received.	All pkts delivered.	Complete

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(b) Selective Repeat (SR)

Event	Sender packet/Action	Receiver/Ack	Window
1	Send pkt 0	Receive & deliver pkt 0, Ack 0	initial window
2	Send pkt 1	Receive & deliver pkt 1, Ack 1	window-advance
3	Send pkt 2	pkt is lost	waiting for Ack 2
4	Send pkt 3	Buffer pkt 3, Ack 3	Receiver base 2
5	Acks 0 & 1 received	Sender advance its base window-advance	
6	Send pkt 4	Buffer pkt 4, Ack 4	Receiver base 2
7	Timeout for pkt 2	Retransmit only Pkt 2	waiting for Ack 2
8	Receive pkt 2	Accept pkt 2, Ack 2	delivered 2, 3, 4 in order
9	Ack 2 received	Sender advance pkt 2, 3 & 4	window-advance
10	Sending remaining pkts as permitted	Receives & Ack pkts	Sliding
11	All packets 0-9 received	All pkts delivered	Complete

Results: GBN retransmit packets 2, 3, 4, giving 3 retransmissions Whereas SR retransmit only packet 2, giving one (1) retransmission.

d) Why counts are different?

Ans: GBN retransmit multiple outstanding packets because it discards out-of-order pkts & uses cumulative acknowledgments whereas SR buffers out-of order pkts & acknowledges them individually, retransmitting only the missing packets.

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Q) Tasks:-

1.

Given:-

R = 2Mbps = 2×10^6 b/s

RTT = 500ms = 0.5s

L = 1000 bytes = 8000bit.

Sol:-

(a) packet transmission time:- $\frac{L}{R} = \frac{8000}{2000.000} = 0.004s = 4ms$.

(b) Stop & wait utilization:-

$U = \frac{L/R}{RTT + L/R} = \frac{0.004}{0.5 + 0.004} = \boxed{0.794\%}$

(c) Utilization for a window of size 4.

$U = \min\left(1, \frac{N(L/R)}{RTT + L/R}\right)$; $U = \frac{4(0.004)}{0.5 + 0.004}$

$\boxed{U = 3.175\%}$

Ans.

Q)

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2.

Part(a).- Sequence numbers available?

$2^k = 2^3 = 8$.

Sequences: 0, 1, 2, 3, 4, 5, 6, 7.

Part(b).- max safe window-size?

$N_{max} = 2^{k-1}$

$= 2^{3-1} = 2^2$

$[N_{max} = 4 \text{ pkts}]$

Part(c):-

Ans: If the SR window-size is greater than half the sequence-number space, a delayed packet from an earlier transmission may have a sequence number that is also valid in the receiver's current window. The receiver could mistake an old duplicate pkt for a new pkt, causing incorrect delivery

Diagram:-

Sequences: 0 1 2 3 4 5 6 7.

Sender: [0, 1, 2, 3, 4] => old-packet (Ack-lost).

Receiver: [5, 6, 7, 0, 1] => Waiting for new.

↑ ↑

Same-number 0 & 1 appear in Both.

Receiver can't tell if it is new or old one.

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